Assignment #2 for Numerical Analysis

Deadline:

1. (30 points) Find all fixed points of the following $g(x)$.
   (a) x^5 (b) $x^2 - 4x + 2$ (c) $\frac{8+2x}{2+x^2}$

2. (30 points) Which of the following three Fixed-Point Iterations converge to $\sqrt{5}$? Rank the ones that converge from fastest to slowest.
   (a) $x \to \frac{4}{5}x + \frac{1}{x}$ (b) $x \to \frac{x}{2} + \frac{5}{2x}$ (c) $x \to \frac{x+5}{x+1}$

3. (30 points) Consider Fixed-Point Iteration applied to $g(x) = 1 - 5x + \frac{15}{2}x^2 - \frac{5}{2}x^3$.
   (a) Show $1 - \sqrt{3/5}$, 1, and $1 + \sqrt{3/5}$ are fixed points.
   (b) Show that none of the three fixed points is locally convergent.

4. (10 points) Show that the initial guesses 0,1 and 2 lead to a fixed point in problem 3. What happens to other initial guesses close to those numbers?